

Juhi Soni

Date of birth: 05/10/1992 **Nationality:** Indian **Gender:** Female

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ABOUT ME

Machine Learning Researcher with 9+ years of experience in computer science, including 4 years specializing in applied machine learning and Generative AI. Experienced in developing automation solutions for industry, with expertise in computer vision, time-series modeling, Agentic AI, and generative models. Skilled at translating advanced research into practical, robust, and performance-driven AI solutions.

WORK EXPERIENCE

MACHINE LEARNING RESEARCH ASSISTANT – FRAUNHOFER IOSB-INA 05/2024 – Current – Lemgo, Germany

- Develop machine learning and Generative AI solutions for real-world industrial applications, translating business and technical requirements into practical AI systems.
- Developed a generative model-based approach for image editing on real-world datasets, improving object
- segmentation accuracy by 3% and reducing manual dataset preparation.
- Developed a BQL query generation framework using an LLM-based agent workflow and RAG, removing the need for employees to have technical knowledge, making data access faster and easier.
- Developed a conditional generative model-based energy disaggregation system to reduce reliance on physical sensors and mitigate data loss caused by damaged or unavailable sensor hardware.
- Developing a machine-learning-based quality inspection system that integrates object segmentation and optical character recognition to automate product inspection, improve efficiency, and reduce manual inspection time.
- Conduct requirement analysis and client discussions, translating business needs into technical requirements and documenting technical solutions and project outcomes.

MACHINE LEARNING WORKING STUDENT – FRAUNHOFER IOSB-INA 12/2021 – 04/2024 – Lemgo, Germany

- Developed and evaluated generative-model-based approaches for anomaly detection in industrial image data, with a focus on reducing reliance on manual data annotation and inspection.
- Built full-stack components for an open-source data-hosting platform using Vue.js and Express.js.

SOFTWARE ENGINEER – GROWEXX 02/2021 – 09/2021 – Ahmedabad, India

- Led full-stack development with Java and AngularJS for enterprise applications in the supply chain domain, conducted unit and integration testing, and worked closely with clients in an agile setting on requirements, user stories, and scrum processes.

SOFTWARE ENGINEER – INNOVIFY 01/2019 – 02/2021 – Ahmedabad, India

- Delivered scalable enterprise solutions using Java and Angular, supporting agile development through requirement analysis, user story definition, and active participation in scrum activities in IOT and Fintech domain.
- Developed backend services and REST APIs to support business workflows and integration between application components.

JAVA DEVELOPER – ZENOSYS SERVICES 02/2017 – 01/2019 – Ahmedabad, India

- Developed Java-based enterprise applications using Spring Boot microservices and event-driven architecture in the supply chain domain.
- Implemented and maintained scalable application components, following software engineering best practices and established architectural patterns.
- Agile development – client communication, requirement analysis, user stories

JOB-RELATED SKILLS

Machine Learning & Generative AI

- Python, Scikit-Learn, Pandas, NumPy, OpenCV
- Statistics, Machine Learning, Deep Learning (PyTorch), CNN, LSTM, Generative Models, VLM
- LLMs, RAG, LangChain, Agentic AI, Prompt Engineering, Vector DB, Hugging Face Transformers, and Diffusers
- Synthetic Data Generation, Anomaly Detection, Time Series Forecasting, Object Detection, Segmentation

ML Systems and Development

- Feature Engineering, Hyperparameter Tuning, Fine-tuning, Model Evaluation, MLflow
- GitHub Copilot, Codex, GPU-based training (NVIDIA DGX)
- Docker, Git, Streamlit

Enterprise Application Development

- Java, Spring Boot, Micro-services, REST APIs, FAST API
- Vue JS, Express JS, Angular JS
- AWS, SQL, Event-driven architecture

Soft Skills

- Research & Critical Thinking, Analytical & Problem Solving, Independent Learning
- Cross-functional Collaboration, Communication & Presentation,
- Technical Documentation, Time and Project Management

PUBLICATION

- Soni J., Lange-Hegermann M., Windmann S. Physics-Informed Diffusion Models for Unsupervised Anomaly Detection in Multivariate Time Series. ECML Workshop on Learning from Small Data, 2025. arxiv.2508.11528.
- Sprute D., Soni J., and Benndorf G. Zero-shot facade parsing for generating building energy modeling inputs using vision-language models and generative image editing. Energy and Buildings Journal, 2026

EDUCATION

M. Sc. – Information Technology – Technische Hochschule Ostwestfalen-Lippe (Grade: 1.6)

20/09/2021 – 24/3/2024 – Lemgo, Germany

- **Thesis:** Diffusion Model for Synthetic Time Series Data Generation from the viewpoint of Physics

Bachelor of Engineering – Bio Medical – Government Engineering College

01/06/2010 – 31/05/2014 – Gandhinagar, India

- **Thesis:** Vowel Training Software

LANGUAGE

- **Gujarati:** Fluent (Mother tongue)
- **English:** Fluent
- **German:** A2 (Currently Learning)